MTH309 PRACTICE PROBLEMS WEEK 4

Question 1. Show that $\mathbb{B}_{\mathbb{R}}$ is generated by the collection $\{(-\infty, r] : r \in \mathbb{Q}\}$, where $\mathbb{Q}$ is the set of rational numbers. Also show that it is not generated by the collection of all the singleton sets or any collection containing a finite number of subsets of $\mathbb{R}$.

Question 2. Let $\mathcal{C}$ be a Monotone class of $\mathbb{R}^d$ containing all the open and closed sets. Show that $\mathbb{B}_{\mathbb{R}^d} \subseteq \mathcal{C}$.

Question 3. Let $\{A_n\}_n$ be a sequence of subsets of Ω. Prove the following statements:

(i) $(\liminf_{n\to\infty} A_n)^c = \limsup_{n\to\infty} A_n^c$.
(ii) $(\limsup_{n\to\infty} A_n)^c = \liminf_{n\to\infty} A_n^c$.
(iii) $\liminf_{n\to\infty} A_n \subseteq \limsup_{n\to\infty} A_n$.
(iv) If $A_n \uparrow A$ or $A_n \downarrow A$, then $\liminf_{n\to\infty} A_n = \limsup_{n\to\infty} A_n = A$.

Question 4. For any subset A of Ω, define the indicator function of A to be $1_A : \Omega \to \mathbb{R}$ given by

$$1_A(\omega) := \begin{cases} 1, \text{ if } \omega \in A, \\ 0, \text{ otherwise.} \end{cases}$$

Let $\{A_n\}_n$ be a sequence of subsets of Ω. For all $\omega \in \Omega$, prove the following statements.

$$\liminf_{n\to\infty} \ 1_{A_n}(\omega) = 1_{\liminf_{n\to\infty} A_n}(\omega),$$

and

$$\limsup_{n\to\infty} \ 1_{A_n}(\omega) = 1_{\limsup_{n\to\infty} A_n}(\omega).$$

Question 5. Let $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ be measurable spaces. Suppose that a collection $\mathcal{C}$ of subsets of Ω_2 generate $\mathcal{F}_2$, i.e. $\sigma(\mathcal{C}) = \mathcal{F}_2$. Then the function $f : (\Omega_1, \mathcal{F}_1) \to (\Omega_2, \mathcal{F}_2)$ is measurable if and only if

$$f^{-1}(A) \in \mathcal{F}_1, \ \forall A \in \mathcal{C}.$$

Question 6. Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be a continuous function. Show that $f : (\mathbb{R}^n, \mathbb{B}_{\mathbb{R}^n}) \to (\mathbb{R}^m, \mathbb{B}_{\mathbb{R}^n})$ is measurable.

Question 7. For any Borel set A in $\mathbb{R}$ and $x \in \mathbb{R}$, define $A + x := \{a + x : a \in A\}$. Show that $A + x$ is a Borel set.

Question 8. Let f and g be real valued Borel measurable functions on $(\Omega, \mathcal{F})$. Fix $A \in \mathcal{F}$ and define the function $h : \Omega \to \mathbb{R}$ by

$$h(\omega) = \begin{cases} f(\omega), \ \omega \in A, \\ g(\omega), \ \omega \in A^c. \end{cases}$$

Show that h is also Borel measurable.

Question 9. (Ash Bk p. 5) Let $A \subseteq \Omega$ and let $\mathcal{F}$ be a σ-field on Ω generated by a collection $\mathcal{C}$. Consider the following collections of subsets of A:

$$\mathcal{C} \cap A := \{B \cap A : B \in \mathcal{C}\},$$

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and

$$\sigma_A(\mathcal{C} \cap A) := \text{the } \sigma\text{-field generated by } \mathcal{C} \cap A \text{ on } A.$$

Prove that $\sigma_A(\mathcal{C} \cap A) = \mathcal{F} \cap A$.
(Hint: To verify this, consider the collection of 'good' sets,

$$\mathcal{E} := \{B \in \mathcal{F} : B \cap A \in \sigma_A(\mathcal{C} \cap A)\}.$$

By the definition of $\mathcal{E}$, $\mathcal{C} \subseteq \mathcal{E}$. Verify that $\mathcal{E}$ is a σ-field. Use the principle of good sets to complete the proof.)